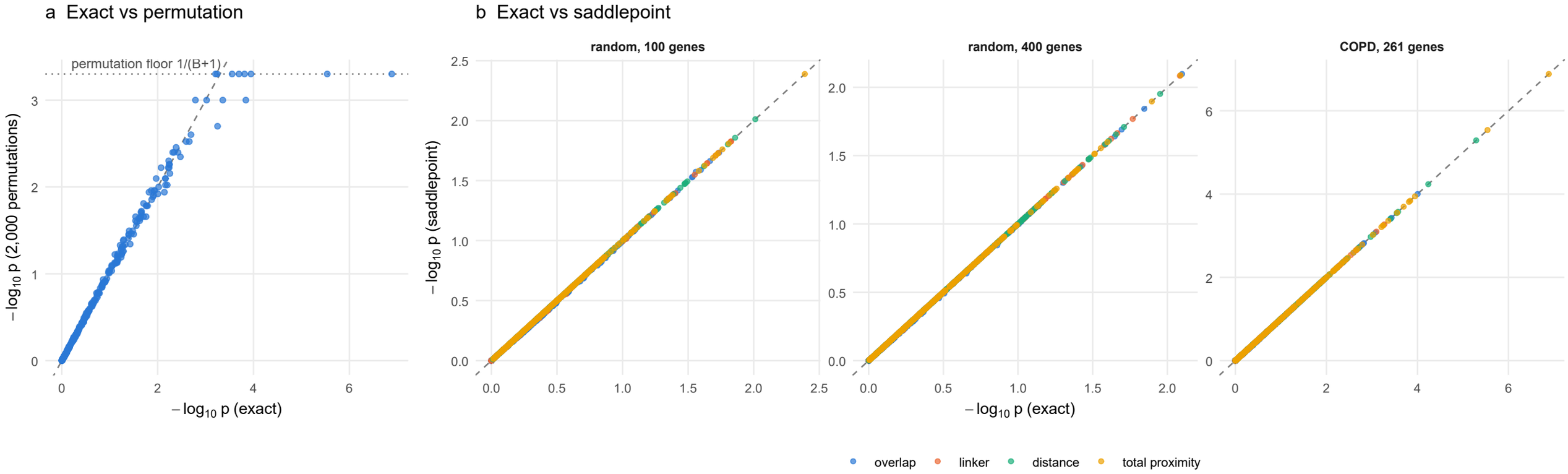

a Exact vs permutation
permutation floor 1/(B+1)
− log10 p (2,000 permutations)
− log10 p (exact)
b Exact vs saddlepoint
random, 100 genes
random, 400 genes
COPD, 261 genes
− log10 p (saddlepoint)
− log10 p (exact)
overlap
linker
distance
total proximity